

ARTICLE

Vegetation Ecology

The presence of American beech litter can alter the growth response of sugar maple seedlings to drought

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Abstract

In late successional forests of North America, sugar maple (*Acer saccharum* Marsh.) and American beech (*Fagus grandifolia* Ehrh.) form a complex ecosystem with intricate interactions. Over the last few decades, several studies have reported a marked increase in American beech dominance relative to sugar maple. Recent evidence suggests that extreme events such as drought could accelerate sugar maple's maladaptation to climate change and favor American beech in its replacement dynamics. In this study, we conducted a greenhouse experiment to investigate the effects of soil water stress and American beech presence on sugar maple seedling growth, structural physiology, leaf nitrogen, and chlorophyll. The seedlings were subjected to the following treatments independently and in combination for 82 days: soil water stress; soil originating from stands with American beech proliferation; soil sterilization; and presence of American beech litter. The results revealed that soil water stress was the primary factor significantly reducing sugar maple seedling growth, which also resulted in an increased root-to-shoot ratio. The presence of soil from stands with American beech proliferation did not exacerbate this negative effect. Soil sterilization, initially expected to reduce seedling growth by eliminating mycorrhizal associations, actually improved seedling growth. This suggests that adverse biotic processes, such as pathogens, were present in the soils regardless of their origin, and their negative effects outweighed the potential benefits from mycorrhization. The addition of American beech litter mitigated the effects of soil water stress but also introduced allelopathic compounds that hindered seedling growth. Overall, this study highlighted the complex interactions affecting sugar maple seedling growth, emphasizing that drought is a major limiting factor.

KEYWORDS

allelopathy, American beech, drought, growth, litter, seedling performance, sugar maple

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INTRODUCTION

Sugar maple (*Acer saccharum* Marsh.) and American beech (*Fagus grandifolia* Ehrh.) are two emblematic species of northeastern North America, playing significant ecological and economic roles as primary sources of wood for the forest industry (Cale et al., 2017; Seymour, 1995). These late successional species form part of a complex ecosystem with intricate interactions (Brisson et al., 1994), where neither species appears to dominate the other (Poulson & Platt, 1996). Yet, in recent decades, several studies have documented a progressive decline in sugar maple populations that is accompanied by an increase in American beech, even resulting in sugar maple regeneration failure in some areas (Beaudet et al., 1999; Bose et al., 2017; Duchesne et al., 2013; Gravel et al., 2011; Hane, 2003; Jenkins, 1997). In sugar maple stands, the proliferation of American beech saplings can lead to the development of dense understories (Bose et al., 2017; Cale et al., 2013), where beech sapling density sometimes exceeds 1500 stems ha⁻¹ (Nolet, Bouffard, et al., 2008). As expansion of American beech over sugar maple has become more commonplace, an extensive literature has emerged to shed light on the interactions between these species, which aim to understand sugar maple decline and to adjust forestry management accordingly (Nolet et al., 2015).

Sugar maple is particularly sensitive to Ca and Mg deficiencies in acidic soil, which hinder its growth and regeneration (Collin, Messier, & Bélanger, 2017; Long et al., 2009; St. Clair et al., 2008). Sugar maple seedlings are also particularly vulnerable to drought. Reduced root colonization by arbuscular mycorrhizal (AM) fungi and root production of seedlings under moisture stress may limit their ability to take up moisture and nutrients (Côté et al., 1998; Liese et al., 2019; Masson et al., 2024). In contrast, sugar maple seedlings compete effectively in light gap areas due to the species' trans-successional status (Canham, 1989; Nolet, Delagrangé, et al., 2008). American beech exhibits high shade tolerance (Beaudet et al., 1999) and tends to create low light environments by regenerating vegetatively through dense thickets, thereby potentially impeding sugar maple survival and growth (Collin, Messier, Kembel, et al., 2017; Dias et al., 2018). Moreover, American beech litter decomposes slowly, resulting in low soil pH and nutrient immobilization in the forest floor (Bélanger & Chaput-Richard, 2023; Côté & Fyles, 1994), together with the release of allelopathic compounds such as phenols through the formation of leachates (Collin, Messier, Kembel, et al., 2017; Hane, 2003). These phenolic compounds can suppress the germination of non-adapted plant seeds and inhibit mycorrhizal symbionts, the latter of which are essential for sugar maple

growth under acidic conditions (Carteron et al., 2020). Thus, American beech not only outcompetes sugar maple on acidic soils (Halman et al., 2015; Kobe et al., 2002), but it can also create unfavorable soil conditions over the long term.

Understanding the interactions between American beech and sugar maple is further complicated by climatic changes affecting forest ecosystem dynamics in recent decades. The increase in extreme events, which may not necessarily incur tree mortality, can subtly alter ecosystem dynamics over long time frames (McDowell et al., 2008; Peng et al., 2011). While the radial growth of sugar maple saplings and mature trees responds similarly to drought (He et al., 2005), seedlings are especially sensitive to soil moisture-deficits due to their restricted root systems (McDowell et al., 2008) with a potential impact on the long-term dynamics of regeneration (Martínez-Vilalta & Lloret, 2016). In a study spanning 57 years of tree growth dynamics, Nolet and Kneeshaw (2018) found that climate change events negatively affected sugar maple growth while American beech remained unaffected. Their results suggested that extreme events like drought could exacerbate sugar maple's maladaptation to climate change and favor American beech in replacement dynamics.

Therefore, the objective of this study was to investigate the factors affecting the growth of sugar maple seedlings in soils that are associated with the proliferation of dense understories of American beech saplings and under extreme events such as drought. We conducted a greenhouse experiment testing the effects of various abiotic and biotic factors on the performance of sugar maple seedlings: soils from stands with American beech proliferation in the understory; presence of American beech litter; availability of microbial symbionts; and soil moisture stress. These factors were assessed both individually and in combination. We hypothesized that seedling performance would be reduced by soil moisture stress, the presence of soil that is conditioned by American beech, the presence of American beech litter, and soil sterilization. We further hypothesized that the negative effects of soil water stress would be amplified when seedlings were growing on soil that was conditioned by American beech and in the presence of American beech litter.

MATERIALS AND METHODS

Study area

We conducted a greenhouse experiment in facilities that were located at the Institut des sciences de la forêt tempérée (ISFORT, Ripon, QC, Canada) from 31 May to

20 August 2021. Soil was sampled from six sites (45.74° to 45.76° N, −74.90° to −74.95° W) that were located within the Kenauk private nature reserve. The climate is cold temperate continental, with average annual air temperature of 5.3°C, ranging from an average minimum of −11.2°C in January to an average maximum of 19.3°C in July (Environment et Changement climatique Canada, 2023, station 7035110, ~7 km from sampling sites). Average annual precipitation is 1204 mm. In the study region, soil water stress (absolute soil water potential ≥ 9000 hPa) lasts an average of 4.5 days, with some areas experiencing durations exceeding 10 days. Projections under a high-emission scenario (RCP8.5) for the 2080 horizon indicate a 33% increase in severity and a 158% increase in duration of soil water stress levels compared with current levels (Cholet et al., 2022). Soils in the study area are classified as Dystric Brunisols (USDA: Typic Dystricceptes), with moder-type humus (FH horizon of ~2 cm), which developed on glacial till deposits that are composed mainly of gneiss, quartzite, and granite (Lajoie, 1967; Soil Classification Working Group, 1998). Soils were collected from 19 to 21 May 2021 in stands that are dominated by sugar maple in the canopy layer with ($n = 3$ sites) or without ($n = 3$ sites) proliferation of dense understories of American beech in the sapling layer. The stands were composed of other companion tree species, such as basswood (*Tilia americana* L.), yellow birch (*Betula alleghaniensis* Britt.), red maple (*Acer rubrum* L.), hophornbeam (*Ostrya virginiana* (Mill.) K. Koch), and eastern hemlock (*Tsuga canadensis* (L.) Carrière). The principal characteristics of each of the two sampled forest environments are summarized in Table 1. At each site (~4000 m²), about 110 (sites without American beech proliferation) to 150 (sites with American beech proliferation) liters of soil (mixture of thin FH layer and 0–15 cm of upper mineral layer) were collected with a soil auger (7 cm diameter). The soil texture of these soils was a sandy loam (4.9% clay and 59.2% sand), with an estimated volumetric water content (VWC) at field capacity of 17.6%.

Experimental design

The greenhouse experiment was comprised of 108 free-draining pots (21.6 cm diameter $\times$ 21.6 cm depth; 6 L) to examine differences in sugar maple seedling response to three soil treatments: soil type (originating from sugar maple stands (1) without proliferation of American beech, (2) with proliferation of American beech, (3) with proliferation of American beech and with the addition of American beech litter), soil sterilization (with vs. without), and soil water stress (with vs. without).

TABLE 1 Characteristics of soil sampling sites within sugar maple stands with and without proliferation of dense understories of American beech saplings.

Characteristics	With American beech proliferation	Without American beech proliferation
Average diameter at breast height (DBH) of trees (DBH > 9 cm) (cm)	22.1	24.5
Stand basal area (DBH > 9 cm) (m ² ha ⁻¹)	29.0	33.1
Basal area proportion of sugar maple (DBH > 9 cm) (%)	84	94
Basal area of American beech in the sapling layer (DBH < 9 cm) (m ² ha ⁻¹)	3.0	0.0
Soil sand content (%)	59.3	59.0
Soil clay content (%)	4.8	5.0
Soil pH	4.8	5.1

Interactions between each soil treatment were also evaluated, resulting in 12 treatment combinations (3 soil types $\times$ 2 soil sterilization types $\times$ 2 soil water stress types), each tested in nine pot replicates (3 pots $\times$ 3 sites).

For each site, half of the soil volume that had been collected was sterilized prior to setting up the experiment using an autoclave (121°C for 2 h), which is recommended to suppress colonization and to depress germination rates and the viability of AM fungi spores (Hu et al., 2020). Each pot was filled with approximately 5 L of soil (54 pots with sterilized soil and 54 with unsterilized soil). We planted a sugar maple seedling in the center of each pot on 31 May 2021. One-year-old seedlings (height = 39 ± 11 cm; mean $\pm$ SD) were obtained as container-grown stock of a common provenance and progeny from the provincial government nursery (Berthierville, QC, Canada). A 3-cm thick layer of American beech litter was deposited on the surface of the soil in the pots that had been subjected to litter addition. This thickness is representative of what can be found in sugar maple stands with beech proliferation as another study conducted in this region recorded an average litter layer of 4.8 cm in stands with beech proliferation, with 57% of the litter mass consisting exclusively of beech litter (A. Maheu, unpublished data). The litter was previously collected in the three sites with beech proliferation. After seedling planting, the 108 pots were randomly distributed in the greenhouse. The experiment included two irrigation phases. The first phase was 15 days (31 May–14 June) of seedling establishment under optimal water moisture

availability conditions for all pots. The second phase of water stress treatment application was 67 days (15 June–20 August), where pots without water stress were watered every 3 days to pot capacity with ca. 450 mL of water and pots with water stress were not watered, until 33% of the seedlings showed signs of leaf wilting (Masson et al., 2024; Pallardy & Rhoads, 1993). These stress conditions were applied three times during the experiment (8 July, 30 July, and 18 August 2017). After each dry spell, the plants that were undergoing water stress were watered to field capacity with ca. 600 mL of water to prevent seedling death. The few weeds were removed manually as they appeared. Seedlings were not fertilized during the pot experiment and only sunlight was used as a light source. The photoperiod fluctuated between 15.4 h on 31 May and 11.9 h on 20 August.

Seedling response variables

Height and basal diameter of seedlings were measured at the beginning and at the end of the experiment. At planting, the seedlings had no branches but produced several over time. The number of branches that each seedling had produced by the end of the experiment was recorded and is referred to as “branching” hereafter. Seedlings were harvested at the end of the experiment (20 August 2021). Aboveground and belowground parts were separated, oven-dried (60°C to constant mass), weighed, and saved for subsequent chemical analysis. Biomass of the aboveground and belowground parts were used to calculate the root/shoot ratio as a means of reflecting biomass allocation by seedlings. Biomass increase was found to be more reliable for analyzing the response of seedlings to treatments than height or basal diameter increase due to high variability in measurement precision. To compute the initial dry mass of seedlings, we developed an allometric model regression equation of the final aboveground biomass (AB, grams of dry matter per tree) as a function of the basal diameter (BD, in millimeters) and stem height (SH, in centimeters): $\log(\text{AB}) = 0.125\text{BD} + 0.014\text{SH} + 0.407$ ($p < 0.001$; adjusted $R^2 = 0.83$). Prediction robustness of the model was then computed on a different subset of seedlings through model II regression comparing estimated values with real observations. The model did not show any sign of bias (slope and intercept did not differ from 1 and 0, respectively) and carried enough precision in the prediction ($p < 0.001$; adjusted $R^2 = 0.819$) to be used for the initial AB calculation. In each pot, the aboveground biomass increment was then calculated as the increase in AB between the beginning and the end of the experiment. The foliar tissues from each plant were

ground; their N concentrations were measured by high-temperature (1450°C) dry-combustion and detection using a TruMac CN analyzer (LECO, St. Joseph, MI, USA). AM colonization was estimated following the protocol described by Parasquive et al. (2023). Briefly, fresh and washed fine roots were selected based upon morphological criteria: absence of lignification, small diameter, and no evidence of necrosis. About 30 root fragments were cleared in 10% m/v KOH, acidified in 1% acetic acid, and stained in 5% black ink-vinegar. Vesicular arbuscular mycorrhizae (VAM) colonization was then determined microscopically (100× magnification) using a gridline intersect method that estimates the frequency of root colonization by arbuscules (Giovannetti & Mosse, 1980). During the experiment, leaf chlorophyll concentration was estimated using a soil plant analysis development (SPAD) chlorophyll meter (502Plus, Konica Minolta, Mississauga, ON, Canada) (three measurements per leaf/three leaves per plant). The SPAD was measured three times on 9 July, 27 July, and 13 August.

Soil response variables

Soil VWC was monitored to 4 cm depth on 11 dates using a FieldScout TDR 300 Soil Moisture Meter (Spectrum Technologies, Aurora, IL, USA); samples were taken 24, 27, and 30 June; 6, 9, 22, 26, and 29 July; and 3, 9, and 12 August. Soil temperature was measured on the same depth and dates using a digital thermometer (Geneq, Montreal, QC, Canada). Before soil sterilization and filling the pots with the soil, soil subsamples from each site were used to determine pH and texture. Bulk pH was measured from deionized water:soil slurries with a ratio of 5:1. Soil texture was determined with the hydrometer method (Bouyoucos, 1962). At the end of the experiment, after seedling harvesting, a soil subsample was collected from each pot to determine pH, together with total C and N that was measured on the TruMac CN analyzer.

Statistical analysis

Data analysis was conducted using the statistical software package R version 4.3.3 (R Core Development Team, 2024). Descriptive statistics were initially calculated to characterize each site. One site exhibited higher acidity and a higher C/N ratio compared with others, irrespective of American beech proliferation. To account for this observed site effect in the experimental design, linear mixed-effects models were used to assess the

significance of differences between measured variables (soil or seedlings) among treatments, with site and potential interactions included as a random factor. For seedling variables that were dependent on height, such as foliage N, the seedling identity was also included as a random variable in the model. Models were computed using the *lme* function of the *nlme* package (Pinheiro et al., 2022). Significant effects underwent further means separation using Tukey's honestly significant difference (HSD) tests, performed with the *glht* function in the *multcomp* package (Hothorn et al., 2008). Biomass increment of the seedlings was based on initial dry mass that was estimated through Model II regression, which was conducted using the *lmodel2* function of the *lmodel2* package (Legendre & Oksanen, 2018). Finally, we used conditional recursive partitioning to identify which applied treatment contributed the most to the seedling biomass increment variation. Regression tree analysis was performed using the *ctree* function of the *party* package (Zeileis et al., 2008).

RESULTS

Soil properties

The soil variables primarily differed between soil types (Table 2). Soils from sites with beech proliferation exhibited slightly, yet significantly, lower pH values and a higher C/N ratio, indicating a lower proportion of soil N compared with sites without American beech proliferation. Soil temperature did not vary between treatments but

fluctuated over the experiment period, ranging from 18.7 to 34.2°C. VWC decreased by 14.5% under induced soil water stress, indicating successful implementation of the treatment. Moreover, VWC was influenced by the addition of American beech litter, increasing by 11.5%. Yet, an interaction between soil water stress and litter treatments was observed (Table 3). Further analysis revealed that while soil water stress reduced VWC, litter addition mitigated this effect, resulting in values similar to soil without litter addition under non-water stress conditions (Figure 1).

Seedling responses to treatment

After 82 days, seedlings in sterilized soils still showed no signs of mycorrhizal colonization, confirming the efficacy of this treatment (Table 4). Mycorrhizal colonization proportion was unaffected by other treatments. Similarly, no differences were observed in seedling structural physiology (i.e., branching) across soil type, sterilization, or soil water stress treatments. Yet, significant differences were noted in foliar N and leaf chlorophyll content through SPAD measurements. Soil sterilization increased foliar N and leaf chlorophyll content, while soil water stress negatively affected leaf chlorophyll content. Interaction between soil type and soil water stress treatments on leaf chlorophyll content (Table 5) revealed that litter addition mitigated the negative effect of soil water stress on chlorophyll content (34.30 ± 0.79 compared with 31.02 ± 0.79 without litter, $p = 0.007$), although still lower than well-watered soils (38.48 ± 0.79 , $p < 0.001$).

TABLE 2 Predicted means (and SEs within parentheses) from the mixed-model analysis of soil variables between treatments (soil type, soil sterilization, and soil water stress) as fixed effects and including site and interactions as random factors.

Treatment	pH		C/N		VWC (%)	
Soil type						
Maple	4.87	(±0.08)b	15.02	(±0.24)c	11.2	(±1.1)b
Maple-Beech	4.59	(±0.04)a	18.86	(±0.17)b	10.6	(±1.2)b
Maple-Beech + litter	4.59	(±0.03)a	19.65	(±0.1)a	22.5	(±2.1)a
Soil sterilization						
No	4.66	(±0.05)	17.92	(±0.31)	15.7	(±1.4)
Yes	4.70	(±0.05)	17.77	(±0.32)	13.8	(±1.4)
Soil water stress						
No	4.61	(±0.05)	17.95	(±0.31)	22.0	(±1.3)a
Yes	4.75	(±0.04)	17.74	(±0.31)	7.5	(±0.5)b

Note: For the soil type treatment, *Maple* refers to soils originating from stands without American beech proliferation, *Maple-Beech* refers to soils originating from stands with American beech proliferation, *Maple-Beech + litter* refers to soils originating from stands with American beech proliferation with the addition of American beech litter. Different letters indicate significant means differences for the treatment.

Abbreviation: VWC, volumetric water content.

TABLE 3 Results of the mixed-model analysis of soil response (pH, C/N, volumetric water content [VWC], and temperature) variables between treatments (soil type, soil sterilization, and soil water stress) as fixed effects and including site as a random factor.

Treatment	df	Statistic	pH	C/N	VWC	Temperature
Soil type	2	χ^2	11.693	256.672	16.224	0.534
		<i>p</i>	0.003	<0.001	<0.001	0.766
Soil sterilization	1	χ^2	1.167	0.109	1.959	1.729
		<i>p</i>	0.28	0.741	0.162	0.189
Soil water stress	1	χ^2	2.731	0.01	35.931	0.102
		<i>p</i>	0.098	0.92	<0.001	0.75
Soil type:Soil sterilization	2	χ^2	0.237	1.953	1.301	0.327
		<i>p</i>	0.888	0.377	0.522	0.849
Soil type: Soil water stress	2	χ^2	3.687	0.846	8.334	0.117
		<i>p</i>	0.158	0.655	0.016	0.943
Soil sterilization:Soil water stress	1	χ^2	0.515	0.387	0.643	0.604
		<i>p</i>	0.473	0.534	0.423	0.437
Soil type:Soil sterilization: Soil water stress	2	χ^2	0.427	1.257	0.478	0.685
		<i>p</i>	0.808	0.533	0.788	0.71

Note: The *p* values are reported for each fixed factor and their interactions. Significant values appear in boldface. χ^2 refers to the statistic value from the Wald χ^2 test.

Seedling growth was primarily affected by soil sterilization or soil water stress (Table 4). Sterilized soils showed significantly higher biomass increment and aboveground biomass investment (lower root-to-shoot ratio), whereas soil water stress had the opposite effect. Strong interactions were observed between these treatments and soil types (Table 5). Soil water stress increased the root-to-shoot ratio significantly, but only for soils originating from sugar maple stands without American beech proliferation (0.57 ± 0.05 compared with 0.87 ± 0.05 in well-watered soils, $p < 0.001$), with no differences in soils with American beech proliferation, either with or without litter addition. Biomass increment was significantly lower with soil water stress, especially on soils without American beech proliferation, further reduced by soil sterilization (Figure 2). This trend was more pronounced in soils with American beech proliferation but remained unaffected by litter addition. Regression tree analysis identified four significant variables contributing to seedling biomass increment (Figure 3). The primary factor influencing seedling growth was soil water stress ($p < 0.001$), followed by litter addition ($p < 0.001$) and soil sterilization ($p \leq 0.025$). The highest biomass increment occurred in well-watered and sterilized soils (node 4, predicted mean of 265.78%), while soil water stress had the most limiting effect on seedling growth (node 11, predicted mean of 45.64%). This negative effect was mitigated by litter addition (node 8, predicted mean of 106.46%) and,

to a lesser extent, by soil sterilization (node 10, predicted mean of 71.37%).

DISCUSSION

This study aimed to explore the effects of factors such as soil originating from sugar maple stands with and without American beech proliferation and climatic events like drought on sugar maple seedling growth. As hypothesized, drought emerged as the primary factor negatively affecting seedling growth through reduced soil water availability. Surprisingly, the presence of soil from stands with American beech proliferation did not exacerbate this effect. Conversely, soil sterilization, initially presumed to be detrimental, actually improved seedling growth. The addition of beech litter interacted both positively and negatively with these effects. Below, we delve into the underlying mechanisms and implications of these findings within natural forest environments in a context of global changes.

Drought was the primary factor reducing sugar maple seedling growth

Our results corroborate expected tree responses to stress, such as drought. Trees in arid climates are typically adapted to invest more biomass into root systems than

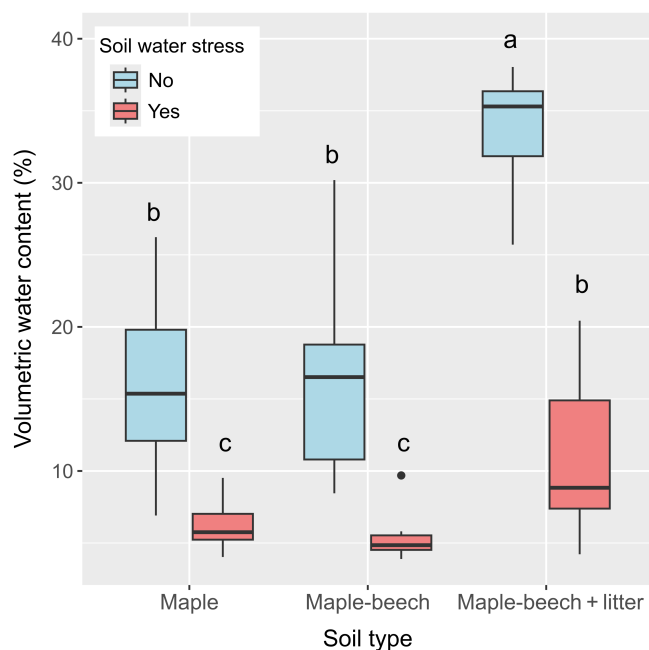


FIGURE 1 Effect of soil water stress and soil type on soil volumetric water content. For the soil type treatment, *Maple* refers to soils originating from stands without American beech proliferation, *Maple-Beech* refers to soils originating from stands with American beech proliferation, *Maple-Beech + litter* refers to soils originating from stands with American beech proliferation with the addition of American beech litter. Different letters indicate significant mean differences.

The lower and upper limits of each box represent the 25th and 75th percentiles, with the line inside the box indicating the median value. Whiskers extend from the box boundaries to the maximum values within 1.5 times the interquartile range (distance between the 25th and 75th percentiles). Data points beyond the whiskers are considered outliers and are plotted individually.

in aboveground parts of the tree to enhance water uptake while minimizing water loss through transpiration (Brunner et al., 2015). This investment shift can also be observed in response to drought events, resulting in an increased root-to-shoot ratio that is dependent on stress intensity (Poorter et al., 2012). Following a drought period, plants tend to reduce water loss through reduced leaf respiration through stomatal closure (Skelton et al., 2017). However, this closure also reduces CO₂ uptake and photosynthetic yield, ultimately hindering tree growth (McDowell et al., 2008). Reductions in tree growth post-drought have been observed in sugar maple trees (Moreau et al., 2020; Nolet & Kneeshaw, 2018; Pilon et al., 1996), as well as in seedlings, together with an increase in root-to-shoot ratio (Ribeyre et al., 2022; present study, Table 4).

Of particular interest is the finding that the addition of American beech litter mitigated the negative effects of the soil water stress treatment on seedling growth (Figure 2), given that litter helped maintain greater soil water content (Figure 1). Tree litter naturally retains water and acts as a buffer against soil desiccation due to evaporation (Walsh & Voigt, 1977). Its removal typically leads to decreased soil water content during dry periods (Sayer, 2006), although recent work suggests that water storage within the litter layer can also reduce soil water recharge (Florincic et al., 2023). While litter cover exists in both sugar maple stands with and without American beech proliferation, our study added American beech litter only to soils from sites with American beech proliferation. Therefore, caution is warranted in drawing conclusions specifically regarding the buffering effects of beech litter on climate changes. Nevertheless, a recent study in sugar maple and American beech forests found that during short-term drought events, American beech litter mitigated decreases in available soil water content, while sugar maple litter did not (Courcot, 2023). This can be explained by the fact that American beech produces recalcitrant litter, which decomposes slowly compared with other deciduous species. For example, 51% of its mass still remains after 6 years (Trofymow et al., 2002), thus ensuring more persistent protection from soil desiccation. In a northern temperate forest study, the decomposition of American beech was found to be three times slower than that of sugar maple (Bélanger & Chaput-Richard, 2023).

Allelopathic effects of American beech litter inhibited sugar maple seedling growth

Sugar maple benefits from microbial associations, such as mycorrhizal symbionts, to cope with adverse conditions like acidic soils or water stress (Carteron et al., 2020; Kannenberg & Phillips, 2017). Microbial associations can also mitigate adverse allelopathic effects that are induced by high levels of phenols in beech litter (Collin, Messier, Kembel, et al., 2017; Hane, 2003). Therefore, we expected soil sterilization to have a more detrimental effect on seedling growth in soils from sites with American beech proliferation (with higher acidity and phenols) or under soil water stress conditions. Surprisingly, sterilization led to increased seedling growth (Figure 2), consistent with findings for multiple temperate tree species, where sterilization increased seedling survival and growth by eliminating soil pathogens (McCarthy-Neumann & Kobe, 2010; Packer & Clay, 2000). While this increase is typically more important for intolerant species compared with shade-tolerant species

TABLE 4 Predicted means (and SEs within parentheses) from the mixed-model analysis of sugar maple seedling response (vesicular arbuscular mycorrhizae [VAM] colonization, branching, foliar N, chlorophyll content measured with a soil plant analysis development (SPAD) meter, root/shoot ratios, and biomass increment) variables between treatments as fixed effects (soil type, soil sterilization, and soil water stress) and including site and interactions as random factors.

Treatment	VAM (%)	Foliar N (%)	Leaf chlorophyll (SPAD value)	Branching (no)	Root/shoot ratio	Biomass increment (%)
Soil type						
Maple	20.56 (±3.98)	2.51 (±0.07)	35.7 (±0.49)	2.72 (±0.27)	0.69 (±0.03)	103.73 (±12.89)
Maple-Beech	18.61 (±3.66)	2.41 (±0.04)	35.2 (±0.61)	2.50 (±0.25)	0.70 (±0.03)	111.09 (±15.67)
Maple-Beech + litter	19.44 (±3.74)	2.48 (±0.06)	36.0 (±0.58)	2.86 (±0.25)	0.71 (±0.03)	122.05 (±9.70)
Soil sterilization						
No	39.07 (±2.16)a	2.38 (±0.04)b	34.4 (±0.35)b	2.48 (±0.19)	0.75 (±0.04)a	85.36 (±8.50)b
Yes	0.00 (±0.00)b	2.55 (±0.05)a	36.8 (±0.53)a	2.91 (±0.22)	0.65 (±0.02)b	139.22 (±11.19)a
Soil water stress						
No	19.72 (±3.23)	2.49 (±0.05)	38.5 (±0.38)a	2.83 (±0.19)	0.66 (±0.02)b	150.09 (±11.93)a
Yes	19.35 (±2.94)	2.44 (±0.05)	32.8 (±0.42)b	2.56 (±0.23)	0.74 (±0.03)a	74.49 (±5.34)b

Note: Seedling identity was also included as a random variable in the foliar N analysis, as foliar N is dependent on seedling height. For the soil type treatment, *Maple* refers to soils originating from stands without American beech proliferation, *Maple-Beech* refers to soils originating from stands with American beech proliferation, *Maple-Beech + litter* refers to soils originating from stands with American beech proliferation with the addition of American beech litter. Different letters indicate significant means differences for the treatment.

TABLE 5 Results of the mixed-model analysis of sugar maple seedling response (vesicular arbuscular mycorrhizae [VAM] colonization, branching, foliar N, chlorophyll content measured with a soil plant analysis development (SPAD) meter, root/shoot ratios, and biomass increment) variables between treatments (soil type, soil sterilization, and soil water stress) as fixed effects and including site as a random factor.

Treatment	df	Statistic	VAM	Branching	Foliar N	Leaf chlorophyll (SPAD)	Root/shoot ratio	Biomass increment
Soil type	2	χ^2	0.503	0.993	2.705	1.683	0.361	4.558
		<i>p</i>	0.778	0.609	0.259	0.431	0.834	0.102
Soil sterilization	1	χ^2	64.991	2.015	4.27	8.239	9.55	25.555
		<i>p</i>	<0.001	0.156	0.039	0.004	0.002	<0.001
Soil water stress	1	χ^2	0.027	0.862	0.222	8.869	4.721	40.086
		<i>p</i>	0.869	0.353	0.638	0.003	0.03	<0.001
Soil type:Soil sterilization	2	χ^2	0.503	1.893	4.027	2.413	1.728	8.002
		<i>p</i>	0.778	0.388	0.134	0.299	0.422	0.018
Soil type:Soil water stress	2	χ^2	0.381	0.717	3.851	9.583	9.84	8.966
		<i>p</i>	0.827	0.699	0.146	0.008	0.007	0.011
Soil sterilization:Soil water stress	1	χ^2	0.027	1.106	0.136	3.646	5.646	4.495
		<i>p</i>	0.869	0.293	0.712	0.056	0.018	0.034
Soil type:Soil sterilization:Soil water stress	2	χ^2	0.381	1.161	2.884	1.681	0.607	3.118
		<i>p</i>	0.827	0.56	0.237	0.432	0.738	0.21

Note: The *p* values are reported for each fixed factor and their interactions. Significant values are reported in bold. χ^2 refers to the statistic value from the Wald χ^2 test.

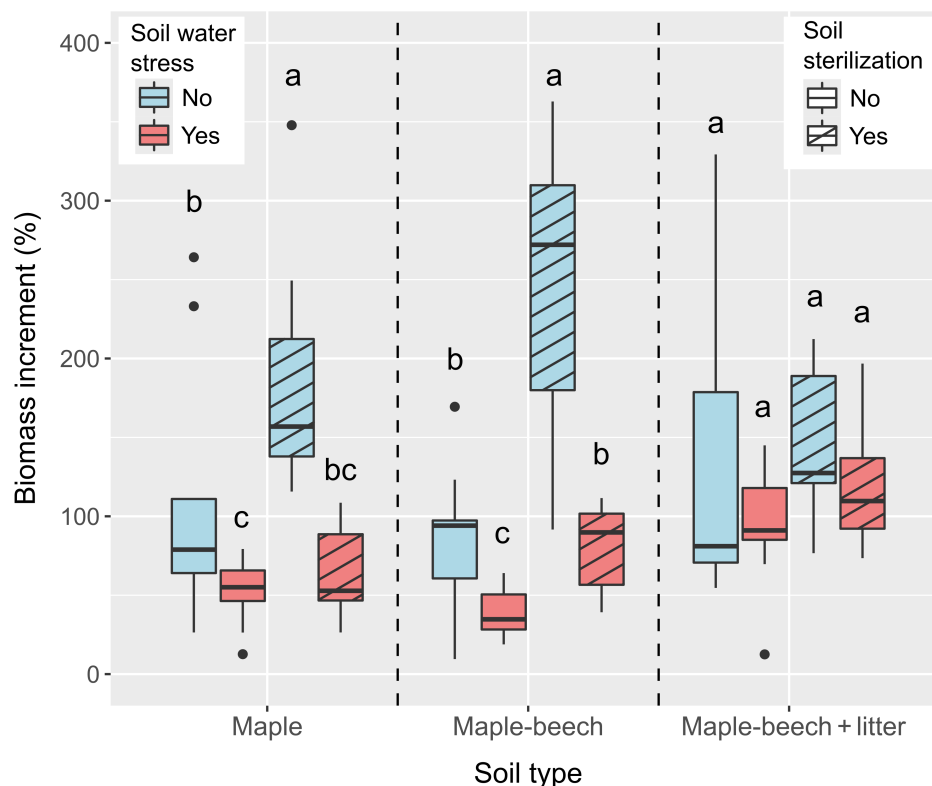


FIGURE 2 Effect of soil water stress, soil sterilization and soil type on biomass increment that was recorded for the sugar maple seedlings at the end of the study. For the soil type treatment, *Maple* refers to soils originating from stands without American beech proliferation, *Maple-Beech* refers to soils originating from stands with American beech proliferation, *Maple-Beech + litter* refers to soils originating from stands with American beech proliferation with the addition of American beech litter. Different letters indicate significant mean differences between treatments within a given soil type. The lower and upper limits of each box represent the 25th and 75th percentiles, with the line inside the box indicating the median value. Whiskers extend from the box boundaries to the maximum values within 1.5 times the interquartile range (distance between the 25th and 75th percentiles). Data points beyond the whiskers are considered outliers and are plotted individually.

like sugar maple (O'Hanlon-Manners & Kotanen, 2006), the presence of soil pathogens can still greatly affect sugar maple seedling growth and survival (Cleavitt et al., 2011). Overall, our results suggest that adverse biotic processes may have been present in the soils regardless of their origin, with their negative effects outweighing potential benefits from mycorrhization. Nonetheless, it should be noted that part of the biomass increase observed with soil sterilization may have been caused by changes in soil nutrient availability. Autoclave sterilization of soil has been linked to the subsequent release of nutrients from the microbial pool and the consumption of organic matter post-sterilization (Hu et al., 2020; Serrasolses et al., 2008). This may have created confounding effects with the positive impact of pathogen elimination, although no differences were detected for soil nitrogen in our study.

Notably, the addition of beech litter appeared to negate the effects of soil sterilization. Adding litter to sterilized soils allowed us to quantify the direct effect of allelopathic compounds that are produced

by American beech litter on sugar maple growth. A meta-analysis of allelopathic effects on plant performance found that allelopathy reduced plant growth by an average of 25%, primarily driven by litter leachate rather than by soil conditioning (Zhang et al., 2021). This aligns with our results, indicating a negative effect of American beech litter on seedling growth but not of soil type, with no difference in seedling growth for soils originating from stands with or without American beech proliferation (Figure 2). Yet, attributing and quantifying reduced growth solely to allelopathy is challenging, given that adding non-sterile litter prevented us from isolating chemical and biotic processes. Microbiota that were present on litter may have mitigated or exacerbated allelopathic effects of phenols on seedling growth (Bonanomi et al., 2021). Additionally, the negative effect of American beech litter was mitigated by the positive effect of soil water retention, explaining why seedling growth was slightly increased in non-sterile soils compared with soils without litter addition.

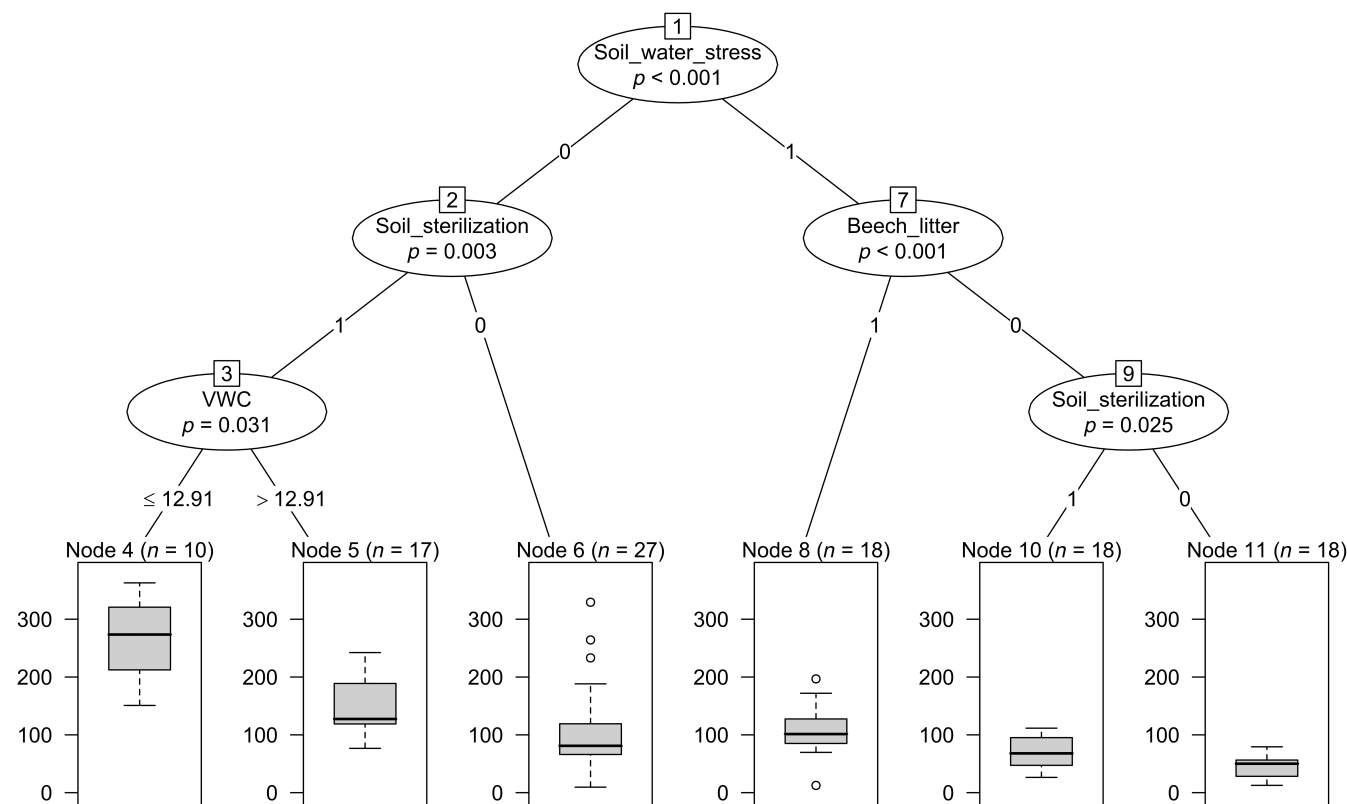


FIGURE 3 Conditional recursive partitioning of sugar maple seedling biomass increment (in percentage) recorded at the end of the study by soil treatments (soil type, soil sterilization, soil water stress, and beech litter addition) and soil variables (pH, C/N, volumetric water content [VWC]). Significant treatments or variables are presented in order of their importance in predicting seedling biomass increment (values shown in the boxplots with n representing the number of seedlings). Threshold values of treatments (1 indicating presence and 0 indicating absence) or variables that lead to specific biomass increments are displayed on the branches of the conditional tree.

CONCLUSION

This greenhouse experiment shed light on the key factors influencing sugar maple seedling growth in forests regardless of American beech proliferation. Proliferation of American beech generally occurs through the development of dense understories, leading to the formation of a thick and persistent litter cover on the forest floor. While this litter cover can mitigate the effects of drought, it is unlikely to benefit sugar maple, given that its regeneration failure is often reported in forests that are affected by American beech proliferation. Conversely, seedlings in forests that are dominated by sugar maple are likely to be more negatively affected by drought due to the thinner litter cover. Complex interactions among climate, soil, and competition will undoubtedly shape sugar maple growth patterns at the local scale. The extent to which these interactions positively or negatively affect sugar maple will depend on the severity and frequency of extreme events such as droughts. This underscores the necessity for further

studies focusing on the interaction between climate, chemical, and biological processes at the local level to better comprehend the trajectories of forest ecosystems on a broader scale.

AUTHOR CONTRIBUTIONS

Alexandre Collin analyzed the data, designed the figures, and wrote the manuscript. David Rivest and Claudele Ghotsa Mekontchou designed the experiment. Claudele Ghotsa Mekontchou and Francesca Sotelo performed the experiment and collected the data. David Rivest, Audrey Maheu, and Phillipe Nolet supervised the project and provided critical feedback to shape the analysis and manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data (Collin, 2024) are available from Zenodo: [10.5281/zenodo.14182973](https://doi.org/10.5281/zenodo.14182973).

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